ModelIt!

Student Activity Pack

# Does More Beavers Mean a Bigger Dam?

How Beaver Builders Shape Their World

Grade NE | NGSS 3-LS4-3, 3-LS4-4, 3-5-ETS1-1 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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# 1 Pre-Assessment

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**What Do You Know?** Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

**Multiple Choice:** Circle the best answer for each question.

**1 What is a dam?**

- **A.** A wall built across a stream that holds water back
- **B.** A type of boat that floats on water
- **C.** A hole dug into the ground to find water
- **D.** A bridge that lets animals cross a river

**2 What do beavers use to build their dams?**

- **A.** Rocks and sand only
- **B.** Trees, sticks, and mud
- **C.** Leaves and grass
- **D.** They do not build anything

**3 What do you think happens to a stream when something blocks the water?**

- **A.** The water disappears completely
- **B.** The water flows faster downstream
- **C.** The water collects and forms a pond
- **D.** The water turns into ice

**4 Why might other animals care about a beaver dam?**

- **A.** Other animals are afraid of beaver dams
- **B.** The pond creates a new place for animals to live
- **C.** Other animals want to destroy the dam
- **D.** Other animals do not notice beaver dams at all

**1 Have you ever seen a beaver or a beaver dam? Where?**

**2 Why do you think beavers build dams?**

**3 Draw what you think a beaver dam looks like from the side.**

**4 What do you think happens to the water when a dam blocks a stream?**

# 2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

# 3 Component Analysis

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### What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

### Sort These Components

Number of Beavers | Trees Available | Dam Size | Pond Size

| EXTERNAL | INTERNAL |
|---|---|
| | |
| | |

**Explain your reasoning: Why did you sort each component the way you did?**

# 4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

**? Explain ONE relationship in your model. What happens when one component changes?**

# 5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

## 1 Scenario 1: Beaver Family

Medium beavers, lots of trees

MY PREDICTION

WHAT I OBSERVED

## 2 Scenario 2: No Trees!

Lots of beavers, no trees

MY PREDICTION

WHAT I OBSERVED

## 3 Scenario 3: Lone Beaver

One beaver, lots of trees

MY PREDICTION

WHAT I OBSERVED

# 6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

## Possible New Components

**Rainfall** — How much rain falls — more rain means more water flowing toward the dam

**Number of Other Animals** — How many fish, frogs, and birds move into the new pond habitat

**Dam Strength** — How well the dam holds up over time — stronger dams last longer

## My Extension

**A Which new component would you add to the model?**

**B How would it connect to the existing components? (Draw or describe)**

**C What evidence or reasoning supports your connection?**

# 7 Reflection Questions

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### The Big Question

**What happens to a pond when beavers build a dam?** Use evidence from your simulation to support your answers.

**1 Why do beavers need BOTH enough family members AND enough trees to build a good dam?**

**2 What would happen to all the animals living in the pond if the beavers left?**

**3 How is what beavers do similar to what human builders do?**

**4 Why do scientists call beavers 'ecosystem engineers'?**

**5 How did your model help you understand something about nature?**

# 8 STEM Challenge: Build-a-Dam Challenge

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## YOUR ENGINEERING MISSION

Design and build a miniature dam using craft sticks and clay that holds back the most water in a plastic bin.

### THE SCENARIO

A nature park wants to understand how beaver dams work. Your engineering team has been hired to build a model dam and test how much water it can hold back. Use what your model taught you about what makes a dam successful!

### Engineering Design Thinking

1. WHAT SHAPE WILL YOUR DAM BE — STRAIGHT, CURVED, OR V-SHAPED?
2. HOW WILL YOU SEAL THE GAPS BETWEEN THE STICKS?
3. HOW WILL YOU MEASURE HOW MUCH WATER YOUR DAM HOLDS BACK?
4. WHAT CAN YOU CHANGE TO MAKE YOUR DAM HOLD MORE WATER?

### My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

**! Explain your design using scientific evidence from the simulation:**

### Real Career Connection

Wildlife Biologists study how animals like beavers change their environments. They work in forests, wetlands, and national parks to protect wildlife. They earn $50,000–$85,000/year.

# 9 Post-Assessment

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## Show What You Learned!

**Multiple Choice:** Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

**1 Why are beavers called 'ecosystem engineers'?**

- **A.** They wear hard hats when they work
- **B.** They change their environment by building dams that create new habitats
- **C.** They are the biggest animals in the forest
- **D.** They only build things for themselves

**2 In the model, what TWO things do beavers need to build a successful dam?**

- **A.** Sunshine and wind
- **B.** Enough beaver family members AND enough trees
- **C.** Rocks and concrete
- **D.** Only lots of water

**3 What happens in the model when there are many beavers but NO trees?**

- **A.** The dam grows very large anyway
- **B.** The beavers use rocks instead
- **C.** The dam cannot grow because beavers need trees as building material
- **D.** The pond gets bigger without a dam

**4 How does dam size affect pond size in the model?**

- **A.** A bigger dam holds back more water, creating a bigger pond
- **B.** Dam size and pond size are not related
- **C.** A bigger dam makes the pond smaller
- **D.** Only rainfall affects pond size